

Ans to the Q. No-1

$$(a) \text{ } \cancel{\text{here}} \quad E(\text{Buys-Computer}) = -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) = 0.94$$

For feature "Student",

$$\cancel{P(\text{yes}) = \frac{6}{7}} \quad P(\text{yes}) = \frac{6}{7}$$

$$P(\text{No}) = \frac{3}{7}$$

$$P(\text{yes}) = \frac{1}{7}$$

$$P(\text{No}) = \frac{4}{7}$$

$$\therefore E(\text{yes}) = -\frac{6}{7} \log_2\left(\frac{6}{7}\right) - \frac{1}{7} \log_2\left(\frac{1}{7}\right) = 0.59$$

$$\therefore E(\text{NO}) = -\frac{3}{7} \log_2\left(\frac{3}{7}\right) - \frac{4}{7} \log_2\left(\frac{4}{7}\right) = 0.98$$

$$\therefore IG(\text{Student}) = E(\text{Buys-Comp}) - \left[P(\text{yes}) * E(\text{yes}) \right] - \left[P(\text{No}) * E(\text{No}) \right]$$

$$= 0.94 - \left[\frac{7}{14} * 0.59 \right] - \left[\frac{7}{14} * 0.98 \right] = 0.155$$

For feature "Credit Rating",

~~$P(\text{Fair}) =$~~

$$P(\text{Yes} / \text{Fair}) = 6/8$$

$$P(\text{No} / \text{Fair}) = 2/8$$

$$P(\text{Yes} / \text{Exc}) = 3/6$$

$$P(\text{No} / \text{Exc}) = 3/6$$

$$\therefore E(\text{Fair}) = -\frac{6}{8} \log_2\left(\frac{6}{8}\right) - \frac{2}{8} \log_2\left(\frac{2}{8}\right) = 0.81$$

$$\therefore E(\text{Exc}) = -\frac{3}{6} \log_2\left(\frac{3}{6}\right) - \frac{3}{6} \log_2\left(\frac{3}{6}\right) = 1$$

$$\therefore IG(\text{Credit Rating}) = E(\text{Buy-Comp}) - \left[P(\text{Fair}) * E(\text{Fair}) \right] - \left[P(\text{Exc}) * E(\text{Exc}) \right]$$

$$= 0.94 - \left[\frac{8}{14} * 0.81 \right] - \left[\frac{6}{14} * 1 \right]$$

$$= 0.048$$

$$\therefore IG(\text{Credit Rating}) < IG(\text{Student})$$

$\therefore$ Student feature will be more suitable.

(Ans.)

~~(b) No~~

(b) No, "unique_ID" ~~is~~ will not be a good feature for classification here. Because "unique_ID" is not actually related with the data points in the ~~D~~ table. This column ~~also~~ has a continuous values ~~which can~~ which are not related to the actual decision. If ~~y~~ we keep that column as a feature, then the ~~model~~ our model will become unable to predict correct values because of the continuous values. It ~~will not~~ can not find any pattern in the datapoints and thus this column would not be a feature.

(c) Yes, we can ~~build a~~ use "Age" as a building node for decision tree.

But at first we need to ~~make some~~ take some steps to handle the continuous values in that column. Here, "Age" column

~~has~~ has many unique values, so we can not calculate the entropy properly. ~~for~~

To do so we need to ~~take~~ take a range of age number ~~for~~ from age values and make them "catagomical". Like, '22-30' is ~~too~~ low-aged, '31-40' is middle-aged and '41-50' is high-aged. Now we can ~~at~~ calculate

the entropy and Information Gain ~~to~~ and then we can use this "Age" feature ~~as~~ as a ~~node~~ building node for decision tree.

$$\begin{aligned}
 (d) \ E(B-C) &= -\frac{6}{14} \log_2\left(\frac{6}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) \\
 &\quad - \frac{3}{14} \log_2\left(\frac{3}{14}\right) \\
 &= \cancel{1.53} \ 1.53
 \end{aligned}$$

for "Income",

$$\begin{aligned}
 \textcircled{2} \quad & \left[\begin{aligned} P(\text{yes}/\text{High}) &= \cancel{2/4} \quad \textcircled{P(\text{No})} \\ P(\text{yes}/\text{Med}) &= 1/6 \\ P(\text{yes}/\text{Low}) &= \cancel{3/4} \end{aligned} \right]
 \end{aligned}$$

$$\left[\begin{aligned} P(\text{No}/\text{High}) &= 2/4 \\ P(\text{No}/\text{Med}) &= 2/6 \\ P(\text{No}/\text{Low}) &= 1/4 \end{aligned} \right]$$

$$\left[\begin{aligned} P(\text{soon}/\text{High}) &= 0/4 \\ P(\text{soon}/\text{Med}) &= 3/6 \\ P(\text{soon}/\text{Low}) &= 0/4 \end{aligned} \right]$$

$$\therefore E(\text{High}) = 1 \quad \left| \quad \therefore E(\text{Low}) = 0.811
 \right.$$

$$\therefore E(\text{Med}) = 1.45$$

Now,

~~250~~

$$\begin{aligned} IG(\text{income}) &= E(B-C) - \left[\cancel{p} P(\text{high}) * E(\text{high}) \right] \\ &\quad - \left[P(\text{low}) * E(\text{low}) \right] \\ &\quad - \left[P(\text{med}) * E(\text{med}) \right] \end{aligned}$$

$$= 0.3911$$

Ans

Ans to the Q.No - 3

Given,

$$P(\text{pos} | \text{malaria}) = 0.94$$

$$P(\neg \text{pos} | \neg \text{malaria}) = 0.04$$

$$P(\text{malaria}) = 0.27$$

~~$P(\neg \text{pos} | \neg \text{malaria})$~~

$$\therefore P(\neg \text{malaria} | \neg \text{pos}) = ?$$

here,

$$~~P(\neg \text{pos})~~ P(\neg \text{malaria}) = 0.73$$

$$P(\neg \text{pos} | \text{malaria}) = ~~0.04~~ 0.06$$

We know,

$$P(\neg \text{malaria} | \neg \text{pos}) = \frac{P(\neg \text{pos} | \neg \text{malaria}) * P(\neg \text{malaria})}{P(\neg \text{pos})}$$

$$= \frac{P(\neg \text{pos} | \neg \text{malaria}) * P(\neg \text{malaria})}{P(\neg \text{pos} | \neg \text{malaria}) * P(\neg \text{malaria})}$$

$$~~P(\neg \text{pos})~~ + P(\neg \text{pos} | \text{malaria}) * P(\text{malaria})$$

$$= \frac{(0.04 * 0.73)}{(0.04 * 0.73) + (0.06 * 0.27)}$$

$$= 0.6431$$

So, probability of the patient actually being malaria negative is 0.6431.

(Ans.)

~~(b) Being~~

(b) Being malaria positive and having a positive test result is ~~actually~~ independent or not we can find that from using the below formula,

~~$$P(A \cap B) = P(A) * P(B)$$~~

~~$$P(\text{pos} \cap \text{malaria}) = P(\text{pos}) * P(\text{malaria})$$~~

$$\therefore \text{L.H.S} = P(\text{pos} \cap \text{malaria})$$

$$= P(\text{pos} / \text{malaria}) * P(\text{malaria})$$

$$= (0.94) * (0.27) = 0.2538$$

$$\therefore \text{R.H.S.} = P(\text{pos}) * P(\text{malaria})$$

$$= [(0.04 * 0.73) + (0.06 * 0.27)] * (0.27)$$

$$= 0.012$$

$$\therefore \text{L.H.S.} \neq \text{R.H.S.}$$

$\therefore$ These events are not independent.

Ans.]

Ans to the Q.No-4

(a) In terms of linear ~~regression~~ regression, ~~the~~

~~the weights, w_1 means~~

the weights, w_1 means the slope of the ~~line~~ line how it is creating a angle with x axis. And the ~~weights~~ weights, w_0 means the y -intercepts of the line.

(b) Given,

$$\begin{array}{l|l} w_1 = 1.5 & \alpha = 1 \\ w_0 = 0 & \end{array}$$

$$\therefore h_w(x) = w_1 x + w_0$$

$$\therefore SSR(h_w) = \sum (y_j - (w_1 x_j + w_0))^2$$

$$= \left[(30 - 1.6w_1 + w_0)^2 + (27 - 2w_1 + w_0)^2 + (24 - 2.5w_1 + w_0)^2 + (22 - 3w_1 + w_0)^2 + (20 - 3.5w_1 + w_0)^2 \right] \quad \text{--- (1)}$$

1st iter 0

$$\frac{d(SSR)(w_1)}{dw_1} = \frac{d}{dw_1} \left[(30 - 1.6w_1)^2 + (27 - 2w_1)^2 + (24 - 2.5w_1)^2 + (22 - 3w_1)^2 + (20 - 3.5w_1)^2 \right]$$

~~$= -493.82$~~

$= \boxed{-493.82}$

$$\text{step size } (w_1) = (-493.82 \times 1)$$
$$= -493.82$$

$$\therefore \text{new } w_1 = \text{old } w_1 - \text{step size}$$

$$= 1.5 + 493.82$$

$$= \boxed{495.32}$$

$$\frac{d(SSR)}{dw_0} = \frac{d}{dw_0} \left[(30 - 1.6 \times 1.5 - w_0)^2 + (27 - 2 \times 1.5 - w_0)^2 + (24 - 2.5 \times 1.5 - w_0)^2 + (22 - 3 \times 1.5 - w_0)^2 + (20 - 3.5 \times 1.5 - w_0)^2 \right]$$

$$= \boxed{-208.2}$$

$$\text{step size} = (w_0) = -208.2 \times 1 \\ = -208.2$$

$$\therefore \text{new } w_0 = \text{old } w_0 - \text{step size} \\ = 0 + 208.2 \\ = \boxed{208.2}$$

2nd itro

$$\frac{d}{dw_1}(\text{SSR}) = \frac{d}{dw_1} \left[(30 - 1.6w_1 - 208.2)^2 + (27 - 2w_1 - 208.2)^2 \right. \\ \left. + (24 - 2.5w_1 - 208.2)^2 + (22 - 3w_1 - 208.2)^2 \right. \\ \left. + (20 - 3.5w_1 - 208.2)^2 \right] \\ = \boxed{38391.83}$$

$$\text{step size} = (38391.83 \times 1) = 38391.83$$

$$\therefore \text{new } w_1 = \text{old } w_1 - \text{step size} \\ = 495.32 - 38391.83 \\ = \boxed{-37896.51}$$

$$\frac{d(SSR)}{dw_0} = \frac{d}{dw_0} \left[(90 - 1 \times 495.32 - w_0)^2 + \right.$$

$$(27 - 2 \times 495.32 - w_0)^2 +$$

$$(24 - 2.5 \times 495.32 - w_0)^2 +$$

$$(22 - 3 \times 495.32 - w_0)^2 +$$

$$(20 - 3.5 \times 495.32 - w_0)^2 \left. \right]$$

$$= \boxed{14318.064}$$

$$\text{step size} = (14318.064 \times 1) = 14318.04$$

$$\Rightarrow \text{new } w_0 = \text{old } w_0 - \text{step size}$$

$$= 208.22$$

$$= 208.22 - 14318.04$$

$$= \boxed{-14109.84}$$

(c) from question 2 we got,

$$w_0 = -14109.84$$

$$w_1 = -37896.51$$

$$\textcircled{Q} \quad x = 3$$

$$\therefore y_{\text{pred}} = -37896.51 \times 3 - 14109.84$$

$$\Rightarrow y_{\text{pred}} = -127799.37$$

✓ So, the y_{act} is 22 for $x=3$ and the

$y_{\text{pred}} = -127799.37$. This ~~pred~~^{pred} predicted value is negative and ~~not~~ the accuracy is bad.

(d) For a large dataset, the learning rates

helps to determine a step size. ~~It is not~~

~~The learning rate is~~ If the learning rate

is decrease in a large data set, then

the step size will be ~~be~~ in change in a very

decreasing rate. So, the step size will be smaller and the iterations will be increasing.

(e) If the learning rate is too small, then in gradient descent algorithm the step size will become smaller.

Because of the smaller step size, the ~~are~~ updated ~~are~~ weights will be update in a ~~decrease~~ small amount. And for that, ~~the~~ iteration number will be increase to decrease the slope.

Ans for Q.No - 2

(a) Initial Domains,

SMT A22

~~SMT 320~~

SMT 251

SMT 4021

SMT 101

SMT 111

& Constraints,

~~cond~~

SMT 251 $\neq$ SMT 321

SMT 101 $\neq$ SMT 360

SMT 422 $\neq$ SMT 421 $\neq$ SMT 320

(b)

~~Req 1: A22~~
Req A:



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Date

(b) using backtracking,

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